

# POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

## VIDEO INFORMATION

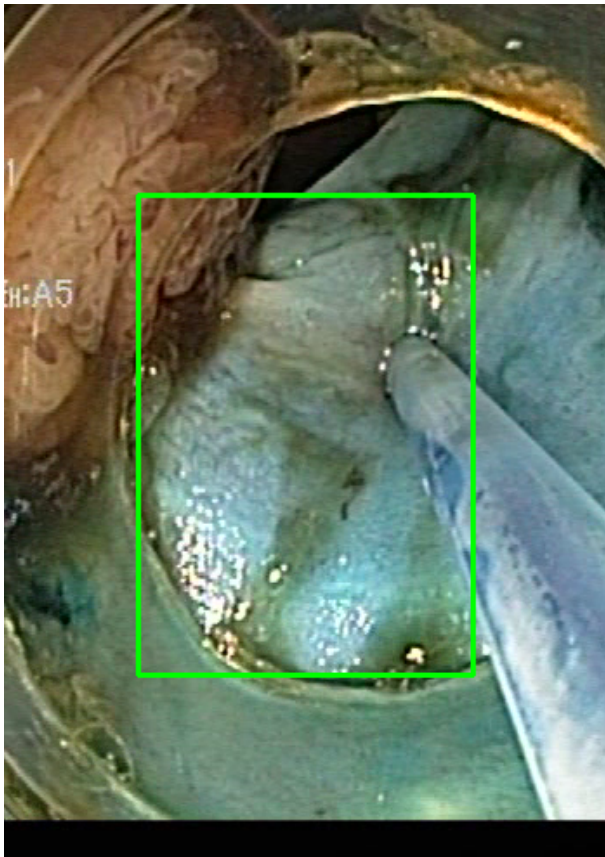
|               |                                      |
|---------------|--------------------------------------|
| Video ID:     | 660fb59a-29ce-4aba-b21c-ae9be157ff2e |
| Total Frames: | 788                                  |
| Frame Rate:   | 25.0 fps (assumed)                   |
| Duration:     | 31.5 seconds                         |

## DETECTION SUMMARY

| Detection Method              | Count        | Status            |
|-------------------------------|--------------|-------------------|
| YOLO Detections               | 18           | ✓                 |
| RT-DETR Detections            | 198          | ✓                 |
| MedSAM2 Detections            | 206          | ✓                 |
| Consensus (3-Model Agreement) | 2            | ✓                 |
|                               |              |                   |
| <b>Risk Classification</b>    | <b>Count</b> | <b>Percentage</b> |
| HIGH RISK                     | 0            | 0.0%              |
| MEDIUM RISK                   | 0            | 0.0%              |
| LOW RISK                      | 0            | 0.0%              |
| TOTAL ANALYZED                | 2            | 100.0%            |

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT\_POLYP [MEDIUM\_RISK] (Tracked across 106 frames)



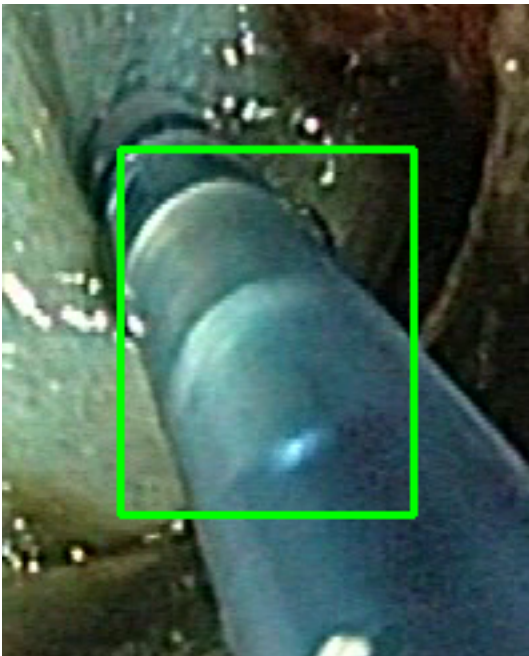
| Feature                 | Value                                | Interpretation                                       |
|-------------------------|--------------------------------------|------------------------------------------------------|
| Frame Index             | 500                                  | Frame 500 of 788                                     |
| Detection Method        | Detector Consensus (MedSAM2 + U-Net) | Model Agreement (MedSAM2 unavailable for this video) |
|                         |                                      |                                                      |
| POLYP TYPE              | FLAT_POLYP                           | Type confidence: 92.0%                               |
|                         |                                      |                                                      |
| Redness Score           | 0.096                                | Color-based vascularization                          |
| Relative Radius         | 33.1%                                | Polyp size as % of frame diagonal                    |
| Texture Score           | 0.929                                | Surface morphology                                   |
| Vessel Visibility       | 0.001                                | Vascularization indicator                            |
| Risk Tier               | MEDIUM RISK                          | Validated by ESGE 2017 / NICE classification         |
|                         |                                      |                                                      |
| Clinical Classification | HIGH_RISK                            | Final symbolic reasoning bucket                      |

|                      |                    |                                                          |
|----------------------|--------------------|----------------------------------------------------------|
| Expert Prediction    | HIGH_RISK          | Decision Tree output                                     |
| Clinical Class       | ADENOMATOUS POLYPS | Polypoid lesion — discrete mucosal protrusion identified |
| Detection Confidence | 67.2%              | YOLO / RT-DETR / track confidence                        |

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — FLAT\_POLYP [MEDIUM\_RISK] (Tracked across 19 frames)



| Feature           | Value                               | Interpretation                                       |
|-------------------|-------------------------------------|------------------------------------------------------|
| Frame Index       | 132                                 | Frame 132 of 788                                     |
| Detection Method  | Detector Consensus (YOLO / RT-DETR) | Model Agreement (MedSAM2 unavailable for this video) |
|                   |                                     |                                                      |
| POLYP TYPE        | FLAT_POLYP                          | Type confidence: 92.0%                               |
|                   |                                     |                                                      |
| Redness Score     | 0.167                               | Color-based vascularization                          |
| Relative Radius   | 7.9%                                | Polyp size as % of frame diagonal                    |
| Texture Score     | 1.000                               | Surface morphology                                   |
| Vessel Visibility | 0.001                               | Vascularization indicator                            |
| Risk Tier         | MEDIUM RISK                         | Validated by ESGE 2017 / NICE classification         |

|                         |                   |                                                          |
|-------------------------|-------------------|----------------------------------------------------------|
|                         |                   |                                                          |
| Clinical Classification | HIGH_RISK         | Final symbolic reasoning bucket                          |
| Expert Prediction       | HIGH_RISK         | Decision Tree output                                     |
| Clinical Class          | ADENOMATOUS_POLYP | Polypoid lesion — discrete mucosal protrusion identified |
| Detection Confidence    | 55.7%             | YOLO / RT-DETR / track confidence                        |

**Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.**

*Low-confidence MODERATE RISK detection - Clinical review recommended*

# CLINICAL IMPRESSION

## Summary:

A total of 2 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

## Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

## Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

## Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

**Disclaimer:** This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.